

Cryptography sheet # 3

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In this sheet, we look again at the invertibility of maps, discuss some more properties of rings and put the extended Euclidean algorithm into practice.

1 Relation of injective, surjective and invertible

Establishing cryptosystems has something to do with inverting functions, since the process of decryption is inversion of a function. Thus, it makes sense to understand the basic properties of invertibility. Let us consider two functions $f: X \rightarrow Y$ and $g: Y \rightarrow X$ acting between sets X and Y such that $g(f(x)) = x$. That is, if you take $x \in X$, then apply f and then apply g , you get x back. Clarify the following:

1. Is f necessarily injective?
2. Is f necessarily surjective?
3. Is g necessarily injective?
4. Is g necessarily surjective?

When the answer is 'yes', explain why, when the answer is 'no' give a counterexample.

2 Some more properties of rings

If you were asked to compute something like $0 \cdot a$ where a is some real number you would not hesitate to answer 0. Similarly, $(-1) \cdot a$ is always equal to $-a$. In some way, elements of commutative rings $(R, +, \cdot)$ behave little bit like numbers. So, one could wonder if the statements from above also hold for them. Try to prove:

1. For all $a \in R$: $a \cdot 0 = 0 \cdot a = 0$.
2. For all $a, b \in R$: $-(a \cdot b) = (-a) \cdot b = a \cdot (-b)$

Bonus exercise: Do we really need R to be commutative?

3 Geometric sums in rings

Most basic math courses teach the so called geometric sum formula, i.e.:

$$\forall n \in \mathbb{N}_0: \quad q \in \mathbb{R} \setminus \{1\}: \quad \sum_{k=0}^n q^k = \frac{q^{n+1} - 1}{q - 1}.$$

It has many applications in real analysis, and its proof is a good exercise in reasoning by induction. So, you are most likely familiar with this formula. But stating it only for real values of q is - if you think about it - pretty restricting. What if someone wants to work with $q \in \mathbb{C}$, $q \in \mathbb{R}^{n \times n}$ or even $q \in \mathbb{Z}_{26}$? Fortunately, it is possible to formulate and prove an analog of the geometric sum formula for any unitary ring. That is what we want to do. Let $(R, +, \cdot)$ be a ring with neutral elements 0_R and 1_R . Then the geometric sum formula can be stated as:

$$\forall n \in \mathbb{N}_0 \quad : \forall q \in R \quad : (q + (-1_R)) \cdot \sum_{k=0}^n q^k = q^{n+1} + (-1_R). \quad (1)$$

Try to prove Equation (1) using induction. Hint: you can use $q^0 = 1_R$. Combining the above formula with the little theorem of Fermat yields another interesting statement. Show that for every prime p we have:

$$\forall q \in \mathbb{Z}_p \setminus \{[1]_p\} : \sum_{k=0}^{p-2} q^k = [0]_p. \quad (2)$$

4 The structure of invertible elements in rings

Let $(R, +, \cdot)$ be any unitary commutative ring. Let G be the subset of invertible elements in R (this set is also called the set of units of R or just $R^\times$). Show that $(G, \cdot)$ is an Abelian group with the respect to multiplication. Remember that you've had the definition of group explained in the previous sheet. You need to verify that the multiplication really gets restricted to an operation $(\cdot) : G \times G \rightarrow G$ ¹ and that the group axioms of Abelian group are true for $(G, \cdot)$.

5 Using the Euclidean algorithm

Now that we have covered the Euclidean algorithm in the lectures, let us put it to practice. Check whether $[15]_{15} \in \mathbb{Z}_{26}$ and $[85]_{2026} \in \mathbb{Z}_{2026}$ are invertible and if so compute there inverses using the extended Euclidean algorithm.

¹i.e., show that $(x \in G \wedge y \in G) \Rightarrow x \cdot y \in G$